Tanyel Bulbul

Ph.D., Associate Professor, Department of Building Construction
Myers Lawson School of Construction, Virginia Tech, Blacksburg, VA, 24160
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Education

2000 - 2006	PhD. in Computational Design , Carnegie Mellon University, Pittsburgh PA Topic: Process and Product Modeling for Computational Support of Building Commissioning.
1997 – 2000	M.Sc. in Architecture METU, Ankara / Turkey
1992 – 1997	Bachelor of Architecture METU, Ankara / Turkey

Academic Appointments

2025 –	Building Construction Program Chair
2022 – 2025	MLSOC Graduate Program Director
2017 –	Associate Professor Department of Building Construction, Virginia Tech
2010 – 2017	Assistant Professor Department of Building Construction, Virginia Tech
2008 – 2010	Post-Doctoral Researcher / Visiting Instructor Computer Integrated Construction Group, Department of Architectural Engineering Pennsylvania State University, State College PA
2008 – 2009	Visiting Instructor Department of Architecture, Pennsylvania State University, State College PA
2006 – 2007	Assistant Professor Istanbul Technical University, Faculty of Architecture
2001 – 2006	Research / Teaching Assistant School of Architecture, Carnegie Mellon University, Pittsburgh PA

Experience

2013 – 2022	Chair and Coordinator, Virtual Design and Construction (VDC) Track, MLSOC Department of Building Construction
2017 – 2020	Faculty Senator, representing MLSOC at the Virginia Tech Faculty Senate
2019 – 2020	Founding Executive Committee Member, representing MLSOC at the College of Architecture and Urban Studies Faculty Association (CAUS-FA)

Funded Research

2025	Coalition for Smart Construction (CSC) “Transforming Precast Concrete Construction through a Cradle-to Cradle Approach”, PI: Tanyel Bulbul (50%), Co-PI: James Jones (25%), Co-PI: Sean McGinnis (25%)

	($75,000)
2024	U.S. Department of Housing and Urban Development (HUD), - subcontract from University of Texas at Austin - "Equitable Technologies for Housing Innovation Center (ETHIC)" Co-PIs: Tanyel Bulbul (50%) and Andrew McCoy (50%) ($337,500) Precast Concrete Institute (PCI) Foundation, "Sustainable Concrete: The Exploration of Reusing Precast Concrete Systems", Co-PI: Tanyel Bulbul (25%), PI: James Jones (50%), Co-PI: Sean McGinnis (25%) COE: Transdisciplinary Research Investments, "Autonomous Construction for Extraterrestrial Environments (ACEE)", PI: Tanyel Bulbul, Co-PIs: Keresh Afsari, Brian Kleiner, Andrew McCoy, Craig Woolsey, Sol Lim, Qin Zhu, Alex Brand ($100,000) COE: SCHEV Equipment Investments, "Real time respirable silica monitors: latest technology to be leveraged with COE's strong industry ties to improve worker health" PI: Emily Sarver, Co-PIs: Tanyel Bulbul and Alex Brand ($18,756.33)
2018	IC-SAFE Granata Pilot Project, "An Assessment of Construction Industry Perceptions of Risk Management Associated with AR Head Display Deployment on Site", PI: Tanyel Bulbul (75%), Co-PI: Joseph Gabbard ($7,000)
2017	U.S. National Science Foundation, "CHS: SMALL: Understanding Human Performance Consequences of Using Headworn Displays for Large Assemblies", PI: Joseph Gabbard, Co-PI Tanyel Bulbul (50%) ($499,984)
2015	ICSafe, Kevin P. Granata Pilot Research Program, "An Interactive App for On-Site Safety Training Based on Surrounding Environment and Types of Work", PI: Tanyel Bulbul (100%) ($4,800) ICTAS, "A Pilot Study for Augmenting Situational Awareness and Real-Time Risk Analysis for Construction Workers through Smart Safety Glasses", PI: Tanyel Bulbul (75%), Co-PI: Joseph Gabbard ($85,570)
2014	Hourigan Construction Company , "Roadmap for AY 2014 - 2015 Coursework and Research". PI: Tanyel Bulbul (75%), Co-PI: Andrew McCoy ($24,940)
2013	Virginia Tech Center for Innovation in Learning, "Digital Construction and Manufacturing". Co-PIs: Tanyel Bulbul (50%), David Goldsmith ($10,000)
2011	Occupational Safety and Health Research Center at VT, Granata Pilot Project "A Pilot Proposal for Assessing the Impact of Regulations on Construction Worker Safety in Building End-of-Lifecycle Operations". PI: Tanyel Bulbul (100%) ($5,000) MLSoC Research Affiliates Program, "Measuring the Impact of BIM and Real-Time Facility Information on Healthcare Environments". PI: Tanyel Bulbul (100%) ($44,698)

2010	Agency for Healthcare Research and Quality, "A Pilot Study for Integrating Facility Information with Healthcare Information to Improve Patient Safety". PI: Chimay Anumba, Co-PIs: Tanyel Bulbul (33%), John Messner. ($100.000)
2009	The Raymond A. Bowers Program, "Applying BIM (Building Information Modeling) Mechanisms to Building End-of-Life Scenarios". PI: Tanyel Bulbul (100%), Co-PIs: John Messner, James Kalsbeek, Tim Baird. ($20.000)

Publications

Refereed Journal Publications

1. Ju, H., Bulbul, T., Yang, X., & Withers, J. (2027). Attention and risk awareness in elevated construction tasks: AR-based simulation with EEG measures. Safety Science, 205, 107428.
2. Ju, H., Bulbul, T., Yang, X., & Withers, J. (2026). Near-real-time EEG-based assessment of perceived attention in AR-enabled construction tasks. Measurement, 283, 122136.
3. Jia, Y., Bulbul, T., McCoy, A. P., & Rausch, C. (2026, in press). Procedure-Aware Contrastive Language–Image Pre-training Model for Construction. Journal of Construction Engineering and Management.
4. Ren, H., Zhang, L., McCoy, A.P., Ganapati, N. E., Bulbul, T., Zhang, R., Whetsell, T. (2026) "Human-Centered Evaluation and Assessment for Resilience (HEAR): Integrating Stakeholder Value Dynamics for Resilience Planning of Residential Buildings." Sustainable Cities and Society (in press).
5. Yang, X., Ju, H., Withers, J., Bulbul, T., (2025, in press) Beyond Surveys: Objective EEG-Based Acceptance of AR-HMDs for Construction Training. ITCon
6. Jia, Y., Bulbul, T., Gabbard, J.L. (2025, in press) An Evaluation of Trade-offs in Augmented Reality Head-Mount Displays for Construction Assembly: Task Performance, Cognitive, and Physical Ergonomics. Journal of Architectural Engineering.
7. Ju, H., Bulbul, T., Yang, X., & Withers, J. (2025, in press). Risk awareness assessment of construction workers wearing AR head-mounted displays using EEG signals. Journal of Safety Research.
8. Ju, H., Bulbul, T., Withers, J. (2025, in press) EEG Signal Cleaning for Brain Wave Monitoring of Construction Workers Using Prediction and Clustering Models. Developments in the Built Environment. https://doi.org/10.1016/j.dibe.2025.100760
9. Jia, Y., Bulbul, T., McCoy, A. P., & Rausch, C. (2025, in press). Assessing cognitive statuses of Hispanic construction workers: A critical review for cognitive ergonomics informed AR/MR development. Journal of Construction Engineering and Management.
10. Yang, X., Jefferson, W., & Bulbul, T. (2025). Evaluation of Construction Worker Perceptions of Wearable Proximity Sensors during the COVID-19 Pandemic. Journal of Construction Engineering and Management, 151(3), 04024217.
11. Qin, Y., & Bulbul, T. (2023). Electroencephalogram-based mental workload prediction for using Augmented Reality head mounted display in construction assembly: A deep learning approach. Automation in Construction, 152, 104892.
12. Qin, Y., & Bulbul, T. (2023). An EEG-based mental workload evaluation for AR head-mounted display use in construction assembly tasks. Journal of Construction Engineering and Management, 149(9), 04023088.
13. Qin, Y., Bloomquist, E., Bulbul, T., & Gabbard, J. (2023). Measuring the impacts of

AR HMD on users' situation awareness during wood frame assembly tasks. Journal of Information Technology in Construction (ITcon), Vol. 28, pg. 70-86, DOI: 10.36680/j.itcon.2023.004

14. Qin, Y., Bloomquist, E., Bulbul, T., Gabbard, J., & Tanous, K. (2021). Impact of information display on worker performance for wood frame wall assembly using AR HMD under different task conditions. Advanced Engineering Informatics, 50, 101423.
15. Yeganeh, A.J., Reichard, G., McCoy, A.P., Bulbul, T., Jazizadeh, F. (2018) Correlation of Ambient Air Temperature and Cognitive Performance: A Systematic Review and Meta-Analysis, Building and Environment, 143, 701-716.
16. Johnston, B., Bulbul, T., Beliveau, Y., Wakefield, R. (2016) "An Assessment of Pictographic Instructions Derived from a Virtual Prototype to Support Construction Assembly Procedures" Automation in Construction, Vol. 64 pp. 36-53
17. Mohammadpour, A., Anumba, C., Bulbul, T., Messner, J., Singh, G., Singh, R. (2015) "Impact Analysis of Facility Failures on the Healthcare Delivery Process: A Use Case Driven Approach" ASCE Journal of Performance of Constructed Facilities (Accepted for publication)
18. Zhao, D., McCoy, A., Bulbul, T., Fiori, C., & Nikkhoo, P. (2015). "Building Collaborative Construction Skills through BIM-integrated Learning Environment." ASC International Journal of Construction Education and Research, Vol. 11/2 pp. 1-24
19. Lucas, J. Bulbul, T. Thabet, W. (2013) "A Conceptual Model of a Proof of Concept Prototype for Healthcare Facility Information Management" ITcon Vol. 18, pg. 76-98
20. Lucas, J. Bulbul, T. Thabet, W. (2013) "An Object-Oriented Model to Support Healthcare Facility Information Management" Automation in Construction 31, 281 – 291
21. Lucas, J. Bulbul, T. Thabet, W., Anumba, C. (2013)"Case Analysis to Identify Information Links between Facility Management and Healthcare Delivery Information in a Hospital Setting" Journal of Architectural Engineering, J. Archit. Eng., 19(2), 134-145
22. Lucas, J., Bulbul, T., Anumba, C. (2013)"Gap Analysis on the Ability of Guidelines and Standards to Support the Performance of Healthcare Facilities" ASCE Journal of Performance of Constructed Facilities, 27(6), 748–755
23. Turkaslan-Bulbul, M.T. Akin, O. (2006) "Computational Support for Building Evaluation: Embedded Commissioning Model" Automation in Construction, Automation in Construction, 15, 438-447.

In Review

1. "A Compliance-Driven Preference Learning Framework for Accessibility Code-Aware Building Layout Generation" Haolan Zhang; Ruichuan Zhang; Kereshmeh Afsari; Lu Zhang; Tanyel Bulbul
2. "A Knowledge-Driven Preference Learning Framework for Accessibility-Aware Building Layout Generation" Haolan Zhang; Ruichuan Zhang; Kereshmeh Afsari; Lu Zhang; Tanyel Bulbul
3. "Assessing A Multimodal Large Language Model for Biophilic Design Evaluation in Sustainable Buildings" Hoda Alemrajabi; Ruichuan Zhang; Tanyel Bulbul
4. Withers, J.W. and Bulbul, T. (2025) Risks that Influence Augmented Reality Head-Mounted Display Usage in the Construction Domain-a Scoping Review. Heliyon, Available at SSRN 4959935.
5. Yang, X., Bulbul, T., Ju, H., Withers, J. (2025) A Balancing Act: MR and EEG-Based Monitoring for Mental Fatigue in Construction Tasks at Height. Journal of Construction Engineering and Management

6. Withers, J.W., Bulbul, T., Ju, H. (2025) Cortical Correlation of Psychophysical Measures to Evaluate Spatial Presence in Augmented Reality Haptic-Based Environments for Construction. Journal of Construction Engineering and Management.
7. Ren, H., Zhang, L., McCoy, A.P., Ganapati, N. E., Bulbul, T., Zhang, R., Whetsell, T. (2025) A Value-Based Framework for Evaluating Alternative Strategies in Community Resilience Planning. – Invited to Lean Construction Journal (LCJ)

In Preparation

8. Hamidi, B., Bulbul, T. (2021) "Object-oriented Data Model to Support Sustainable Demolition Waste Management Decision Making at End-of-Life"
9. Hamidi, B., Bulbul, T. (2021) "A Framework for a Proof of Concept Prototype for Building End-of-Life Decision Making"
10. Wang, Z., Bulbul, T., McCoy, A.P. "A Comparison of Facility Management Data to Building Handover Documents: Philip Merrill Center Case Study". Journal paper.
11. Wang, Z., McCoy, A.P. Bulbul, T., "Innovating for Virtual Construction Management: Assessing the Benefits of Field Data for Turnover and Property Management".

Book Chapters and Monographs

1. Bulbul, T. (2012) "Product and Process Models for Building Commissioning" Embedded Commissioning of Building Systems: Evaluating Buildings in the Information Age, ed. Omer Akin et al. Artech House, Boston.
2. Bulbul, T. (2012) "IFC, STEP: Standards " Embedded Commissioning of Building Systems: Evaluating Buildings in the Information Age, ed. Omer Akin et al. Artech House, Boston.
3. Lucas, J., Bulbul, T., (2015) "Ontology to Support Healthcare Facility Management" ASCE Monograph, Ontology in Civil Engineering: A Decade of Research and Development, ed. Raymond Issa

Refereed Conference Publications

2026	1. Zhang, H., Zhang, R., Afsari, K., Zhang, L., & Bulbul, T. (2026). Evacuation Performance-Aware Building Layout Generation for Early-Stage Architectural Design. ASCE International Conference on Computing in Civil Engineering, Incheon, South Korea, 14 - 17 June. 2. Kia, K., Ayer, Y., Afsari, K., Bulbul, T. (2026) In-Situ Resource Utilization (ISRU) for Autonomous Remote Construction: Trends and Applications, ASCE Construction Research Conference, San Antonio, TX, 18 - 21 March 2026. 3. Zhang, R., Mallepally, A., Bulbul, T. (2026) Generative AI and Augmented Reality for Personalized Biophilic Design in Senior Living Environments, ASCE Construction Research Conference, San Antonio, TX, 18 - 21 March 2026. 4. Alemrajabi, H., Zhang, R., Bulbul, T. (2026) Exploring the Utility of Large Language Models (ChatGPT) in Biophilic Design Evaluation, ASCE Construction Research Conference, San Antonio, TX, 18 - 21 March 2026.
2025	5. Alemrajabi, H., Zhang, R., Bulbul, T., Pearce, A. (2025) Integrating Nature into Norms: Evaluating Biophilic Design Attributes in Building Rating Systems. CIB World Building Congress WBC2025, West Lafayette, Indiana. 6. Alemrajabi, H., Zhang, R., Bulbul, T., Gao, X., Agee, P. (2025) Bridging Nature and Architecture: Generative AI for Assessing Biophilic Design Attributes. ASCE International Conference on Computing in Civil Engineering, New Orleans, Louisiana, 11 - 14 May

	7. Ren, H., Zhang, L., McCoy, A.P., Ganapati, N. E., Bulbul, T., Zhang, R., Whetsell, T. (2025) A Value-Based Framework for Evaluating Alternative Strategies in Community Resilience Planning. ASCE Construction Research Congress (CRC), Montreal, Canada, 28-31 July
2024	8. Yang, X., Bulbul, T., Qin, Y. (2024) Current State of the Art in Real-Time EEG-Based Recognition of Mental Status in Construction, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 28 - 31 July. 9. Withers, J., Bulbul, T. (2024) Towards a Conceptual Model to Measure the Influence of AR-HMDs on Workers' Perception in Construction Environments Withe, ASCE International Conference on Computing in Civil Engineering, Pittsburgh, Pennsylvania, 28 - 31 July.
2023	10. Qin, Y., Bulbul, T., Withers, J. (2023) EEG-based Classification of Cognitive Load and Task Conditions for AR Supported Construction Assembly: A deep learning approach, Computing in Civil Engineering, Corvallis, Oregon, 25 - 28 June. 11. Budzevski, L., Surana, N., Bulbul, T., Zhang, R. (2023) Augmented Reality and Wearable Technology Supported Biophilic Design of Senior Housing for Improving Quality of Life in Older Adults, Computing in Civil Engineering, Corvallis, Oregon, 25 - 28 June.
2022	12. Qin, Y., Bulbul, T. (2022) Measuring the Impact of Information Display Methods on AR HMD for Comprehending Construction Information with EEG, ASCE Construction Research Conference, Arlington, Virginia, 9 - 12 March 2020, pp. 235-243.
2020	13. Qin, Y., Bulbul, T. (2020) Towards An EEG Based Mental Workload Evaluation Method for Construction Worker's HMD AR Use, ASCE Construction Research Conference, Tempe, Arizona, 8 - 10 March 2020, pp. 1105-1109. 14. Bloomquist, E., Gabbard, JL,Tanous, K., Qin, Y. Bulbul, T. (2020) Framing the Scene: An Examination of Augmented Reality Head Worn Displays for Construction Assembly Tasks, IEEE Virtual Reality, Atlanta, GA, 22 - 26 March 2020.
2018	15. Akanmu, A., Ojelade A., Bulbul, T. (2018) Gaming Approach to Designing for Maintainability: A Light Fixture Example, 35th International Symposium on Automation and Robotics in Construction and Mining (ISARC 2018) and the International AEC/FM Hackathon, Berlin, Germany, 20 - 25 July 2018, pp. 1105-1109. 16. Akanmu, A., Ojelade A., Bulbul, T. (2018) Reassigning Construction Laborers based on Body Motion Analysis, 35th International Symposium on Automation and Robotics in Construction and Mining (ISARC 2018) and the International AEC/FM Hackathon, Berlin, Germany, 20 - 25 July 2018, pp. 1098-1105. 17. Xiiong, Y, Bulbul, T., Reichard, G. (2018) BIM and Game Engine Integration for Operational Data Monitoring in Buildings, 7th International Building Physics Conference, IBPC2018, Syracuse, NY, September 23-26 2018
2016	18. Bulbul, T. (January, 2016) "Using Pictographic Instructions Derived from a Virtual Prototype to Support Construction Tasks." VT CAUS Bi-Annual Research Symposium, Blacksburg, VA.
2015	19. Wang, Z., Bulbul, T., McCoy, A. (2015) "A Comparative Case Study for Using COBie in Project Turnover" CIB W078 Computing Conference, October 2015, Eindhoven, The Netherlands.

	20. Wang, Z., Bulbul, T., Lucas, J. (2015) “A Case Study of BIM-Based Model Adaptation for Healthcare Facility Management - Information Needs Analysis” Proceedings of the ICCCBE/ASCE/CIB W078 Computing Conference, June 2015, Austin, TX.
2014	21. Chen, J., Bulbul, T., Taylor, JE., Olgun, G. (2014) “A Case Study of Embedding Real Time Infrastructure Sensor Data to BIM” Proceedings of the Construction Research Congress, May 2014, Atlanta, GA. 22. Hamidi, B., Bulbul, T., Pearce, A., Thabet, W. (2014) “Potential Application of BIM in Cost-benefit Analysis of Demolition Waste Management” Proceedings of the Construction Research Congress, May 2014, Atlanta, GA. 23. Muthumanickam, A., Rishee, J., Taylor, JE., Bulbul, T. (2014) “Development of a Novel BIM-Energy Use Ontology” Proceedings of the Construction Research Congress, May 2014, Atlanta, GA. 24. Hamidi, B., Bulbul, T. (2014) “Environmental Impacts Analysis of End of Building Life Cycle Operations” Proceedings of the ICCCBE/ASCE/CIB W078 Computing Conference, June 2014, Orlando, FL. 25. Mohammadi, N., Bulbul, T., Taylor, JE. (2014) “Evaluating Current Systems and Exploring the Potential of Social Networking Platforms to Increase User Engagement with Eco-Feedback Systems” Proceedings of the ICCCBE/ASCE/CIB W078 Computing Conference, June 2014, Orlando, FL. 26. Short, K.M., Pearce, A.R., Fiori, C.M., Bulbul, T., and McCoy, A. (2014). “The Influence of Internship Participation on Construction Industry Hiring Professionals When Selecting New Hires and Determining Starting Salaries for Construction Engineering Graduates.” Conference Proceedings, 121st ASEE Annual Conference and Exposition, ASEE, Indianapolis, IN.
2013	27. Lucas, J. Bulbul, T. (2013) "Future Directions for a Healthcare Facility Information Management System" in Proceedings of the ASC 49th Annual International Conference, San Luis Obispo, California
2012	28. Mohammadpour, A., Anumba, C., Bulbul, T., Messner, J. (2012) “Facilities Management Interaction with Healthcare Delivery Process” Proceedings of the Construction Research Congress, May 2012, West Lafayette, IN. 29. Hamidi, B. Bulbul, T. (2012) "A Comparative Analysis of Sustainable Approaches to Building End-of-Lifecycle: Underlying Deconstruction Principles in Theory and Practice" in Proceedings of the ASCE International Conference of Sustainability in Design, Engineering and Construction (ICSDEC), W.K.O. Chong, J. Gong, J. Chang, M.K. Siddiqui Editors, ASCE, Reston, Virginia, pp. 155-162.
2011	30. Lucas, J., Bulbul, T., Anumba, C., Messner, J. (2011) “Evaluating the Role of Healthcare Facility Information on Health Information Technology Initiatives from Patient Safety Perspective” in Proceedings of the ASCE International Workshop on Computing in Civil Engineering, Y. Zhu, R.R. Issa, Editors, ASCE, Reston, Virginia, pp. 720-727. 31. Bulbul, T. (2011) “An Analysis of Construction Worker Safety during Building Decommissioning and Deconstruction” Proceedings of the CIB W099 Safety and Health in Construction Conference, August 2011, Washington, DC. 32. Lucas, J. Bulbul, T. Thabet, W. (2011) “A Facility Information Model to Support Operations and Maintenance of a Healthcare Environment” Proceedings of the CIB

	W078, W102 Computer Knowledge Building Conference, October 2011, Sophia Antipolis, France.
2009	33. Bulbul, M.T. Anumba, C. Messner, J. (2009) "A System of Systems Approach to Intelligent Construction Systems" in Proceedings of the 2009 ASCE International Workshop on Computing in Civil Engineering, C.H. Caldas, W.J. O'Brien, Editors, ASCE, Reston, Virginia, pp. 22-32.
2008	34. Gul, F.G. Wang, X. Turkaslan-Bulbul, M.T. Cagdas, G. Tong, H. (2008) "Global Teamwork: A Study of Design Learning in Collaborative Environments" DRS2008, Design Research Society Biennial Conference, Sheffield, UK.
2007	35. Lee, S. Turkaslan-Bulbul, M.T. Akin, O. (2007) "The Development of an Augmented Reality-Based Data Visualization System for Commissioning of Air Handling Units", in Proceedings of the 7th International Conference on Construction Applications of Virtual Reality, J. Messner, Editor, University Park, Pennsylvania, pp. 155-164. 36. Turkaslan-Bulbul, M.T. Akin, O. (2007) "Evaluation Methods for Building Product Models: Measuring the Performance of Building Commissioning Data Model", in Proceeding of the 2007 ASCE International Workshop on Computing in Civil Engineering, L. Soibelman, B. Akinci, ASCE, Reston, Virginia, pp. 272-280.
2005	37. Turkaslan-Bulbul, M.T. Akin, O. (2005) "Formal Models for Embedded Commissioning", Proceedings of the International Conference for Enhanced Building Operations, Pittsburgh, Pennsylvania 38. Turkaslan-Bulbul, M.T. Akin, O. (2005) "Computational Support for Building Evaluation: Embedded Commissioning Model", in The Future of the AEC Industry, C. Eastman, T. Galloway, R. Kangari, G. Lee, L. Thomas-Mobley, Building Futures Council, Las Vegas, Nevada, pp. 178-188 39. Turkaslan-Bulbul, M.T. Akin, O. (2005). "A Review of Building Product Models in AEC Industry Towards a Model for Building Commissioning". in Innovation in Architecture, Engineering and Construction, S. Sariyildiz, B. Tuncer, Editors, SOON, Rotterdam, The Netherlands.
2004	40. Wang, H., Akinci, B., Garrett, J.H., Akin, O., Turkaslan-Bulbul, M.T. and Gursel, I. (2004). Towards Domain-Oriented Semi-Automated Model Matching for Supporting Data Exchange. Proceedings of the 10th International Conference on Computing in Civil and Building Engineering, Weimar, Germany. 41. Akin, O. Turkaslan-Bulbul, M.T. Gursel, I. Garrett, Jr. J.H. Akinci, B. Wang H. (2004) "Embedded Commissioning for Building Design" Proceedings of ICEBO Conference, Paris, France. 42. Akin, O. Turkaslan-Bulbul, M.T. Gursel, I. Garrett, Jr. J.H. Akinci, B. Wang H. (2004). "Embedded Commissioning for Building Design". in ECPPM 2004 eWork and eBusiness in Architecture, Engineering and Construction, A. Dikbas, R. Scherer, Editors, Taylor & Francis Group, London, UK, pp. 343-357
2003	43. Akin, O. Turkaslan-Bulbul, M.T. Brown, S. Kim, E. Akinci, B. Garrett, Jr. J.H. (2003). "Comparison of ASHRAE Guidelines with Building Commissioning Practice"

	Proceedings of National Conference on Building Commissioning, Palm Springs, California. 44. Turkaslan-Bulbul, M.T. Akin, O. Garrett, Jr. J.H. Akinci B. (2003). "Overview of Design Evaluation towards a Comprehensive Computational Approach" " in E-Activities in Design and Design Education, B. Tuncer, SS. Ozsariyildis and S. Sariyildiz, Editors, Eurapia Productions, Paris, France, pp. 203-209.

Honors & Awards

2023 - 2024	Deans Fellow, College of Engineering, Virginia Tech
2016	Virginia Tech, College of Architecture and Urban Studies (CAUS) Diversity Recognition Award This award is given for my involvement with the Building Women in Construction (BWIC) as the faculty advisor and her commitment to STEM outreach with girls in local elementary and middle school.
2014	New Media Seminar, Sponsored by Virginia Tech
2012	Faculty Fellowship award from Balfour Beatty Construction to observe renovation of OR suites at Walter Reed national Military Medical Center at Washington, DC.
2011	Proposal Development Institute, Sponsored by Virginia Tech
2010	US-Nigeria International Workshop on Innovative Design Approaches to Sustainable Buildings, Sponsored by NSF
2010	ASCE ExCEEd (Excellence in Civil Engineering Education) Workshop, Sponsored by Department of Building Construction
2007	Academic Exchange, Technical University of Delft, Sponsored by European Union, Socrates-Erasmus Program
2005	Building Futures Council, Graduate Student Conference sponsorship to attend The First Conference on the Future of the AEC Industry, Las Vegas, Nevada ***This paper was selected as one of the innovative ideas that will affect the future of Architecture/Engineering/Construction industry*** March 2005
2003 - 2006	Full scholarship from Carnegie Mellon University to pursue a PhD
2000 - 2004	Full scholarship from Turkish Higher Education Council to pursue a PhD

Academic Service

2020 - 2022	Committee member, MLSOC Curriculum Committee
2020	Committee member, CAUS Associate Dean of Research Search Committee
2019	Committee member, CAUS Interior Design Faculty Search Committee
2019 - present	Committee member MLSOC Promotion and Tenure (P&T) Committee
2018	Internal Peer Reviewer for Virginia Tech's Academic Program Review (Department of Finance, Insurance, and Business Law)
2019 - present	Faculty Advisor, FIRST Alumni Association student organization
2018 - 2020	Committee member (Faculty Senate rep) _ Virginia Tech Campus Development Committee
2017 - 2019	Committee member (CAUS rep) _ Virginia Tech Commencement Committee
2017 - 2018	Committee member, MLSOC Diversity Committee
2017	Committee member, CAUS Dean Search Committee

2017	Committee member, MLSOC Faculty Search Committee
2015 - present	Editorial Review Board member _ Engineering, Construction and Architectural Management (ECAM)
2015 July	Specialty Editor, Buildings _ "BIM in Building Lifecycle"
2014 - present	Faculty Advisor, Building Women in Construction student organization
2014 - 2017	Committee member (College of Architecture and Urban Studies rep) _ Virginia Tech Graduate Curriculum Committee
2014	"From Design to Construction and Contraction to Facility Management: BIM and Building Lifecycle" Panel Organizer, ASC 50th Annual International Conference, March 2014
2010 - 2014	Committee member _ Virginia Tech College of Architecture and Urban Studies Honorifics committee
2010 - 2014	Committee member _ Virginia Tech College of Architecture and Urban Studies Diversity committee
2010 – present	Committee member _ ASCE Global Center of Computing Excellence This group recognizes that global challenges in computing and civil engineering cannot be met by the efforts of one country, and it aims to support the development of civil engineering-specific computing knowledge, sharing teaching modules and standards.
2009 – present	Committee member _ ASCE Data Sensing and Analysis Committee The group is composed of international researchers and it aims to gather, maintain and disseminate information on the application of expert systems and artificial intelligence to civil engineering, and to keep the Society membership aware of developments in this rapidly growing field.
2009 – present	Committee member _ ASCE Visualization, Information Modeling, and Simulation Committee The purpose of this international committee is to increase the appropriate use of databases and information management technologies in civil engineering teaching and practice.
2007	"Reflective: Animating Urban Surfaces, Programming + Design" Workshop organized together with Gulen Cagdas, Ahu Sokmenoglu and Esra Gurbuz from Istanbul Technical University and Mona Mahall and Asli Serbest from University of Stuttgart, March 2007
2005	"Collecting, Modeling, Exchanging and Visualizing Building Commissioning Information Relevant to First Responders" Workshop Coordinator, International Conference for Enhanced Building Operations, October 2005
2004	Technical assistant, reviewer and symposium coordinator in International Symposium on Generative CAD: G-CADS July 2004
2003	Graduate Student Representative in the Graduate Student Assembly of the Carnegie Mellon University from 2003 to 2004 (elected by fellow students).
2002	Computational Design Student Representative in the MS/PhD. committee of the School of Architecture at CMU 2002 to 2003 (elected by fellow students).

Advised Students

Doctoral

- Jason Lucas (Co-Chair with Walid Thabet)

Title: “An Integrated BIM Framework to Support Facility Management in Healthcare Environments”. Graduated December 2012. Dr Lucas is an Associate Professor at the Clemson University.

- Behzad Hamidi (Chair)
  Title: “A BIM-based Object-oriented Data Model to Support Sustainable Demolition Waste Management”. Graduated May 2015. Dr Hamidi is a project manager at USPS.
- Yimin Qin (Chair)
  Title: “A Physiological Evaluation of Construction Worker’s Mental Workload for HMD AR Use”. Graduated May 2023
- Jeremy Whiters (Chair)
  Title: “Identifying Deployment Risks for Augmented Reality Head Wearable Displays in the Construction Industry”. May 2026
- Hoda Alemrajabi (Co-Chair with Ruichuan Zhang)
  Title: “Exploring the Incorporation of Biophilic Design in Built Environment Using Generative AI”. Expected graduation December 2026

- Xiaoying Yang (Chair)
  Title: TBD. Expected Graduation May 2027
- Yuhang Jia (Chair)
  Title: TBD. Expected Graduation December 2026
- Hanwen Ju (Chair)
  Title: TBD. Expected Graduation May 2027

- Leeann Budzevski (Co-Chair with Ruichuan Zhang)
  Title: TBD. Expected Graduation May 2026

- Han Reng (Committee Member)
  Title: Integrating Stakeholder Value Systems into Disaster Resilience Planning: A Value-Driven Framework for Collaborative Decision Making
- Ping Xu (Committee Member)
  Title: A Framework for Creating Digital Twins for Robotic Indoor Operations: A Method Integrating BIM, IoT, Graph Theory, and Computer Vision
- Haolan Zhang (Committee Member)
  Title: TBD

- Yunjie Xiong (Committee Member)
  Title: “A BIM-based Interoperability Platform in Support of Building Operation and Energy Management”. Graduated February 2020
- Eric Wetzel (Committee Member)
  Title: “The Use of a BIM-Based Framework to Support Safe Facility Management Processes”. Graduated December 2018
- Brendan Johnston (Committee Member).
  Title: "A Graphical Proof for Virtual Prototype Constructability Translation." Graduated 2012
- Kathleen Short (Committee Member).
  Title: "Assessing the Outcomes of Internship Experiences Utilizing the SLICES model for Students Entering the Construction Industry." Graduated May 2013
- Kongkun Charoenvisal (Committee Member).

Title: “A BIM Interoperable Web-Based DSS for Vegetated Roofing System Selection” Graduated August 2013

- Alan M. Lytle (Committee Member).
  Title: “A Framework for Object Recognition in Construction Using Building Information Modeling and High Frame-Rate 3D Imaging”. Graduated May 2011

Master of Science

- Will Wang (Chair)
  Title: TBD. Expected Graduation May 2025
- William Jefferson (Chair)
  Title: “Construction Worker Perceptions of Wearable Proximity Sensors” Graduated May 2021
- Zhulin Wang (Chair)
  Title: “BIM-Based Personnel Training Modules for Healthcare Facility Management”. Graduated 2019
- Parag Garg (Chair)
  Title: “Application of Augmented Reality Using Head Mounted Display in Construction”. Graduated December 2018
- Irene Soto (Chair)
  Title: “A Pilot Test for Identifying Tasks and Degrees of Visual Fidelity for Applications of Head Mounted Display Systems (HMD) for Construction”. Graduated August 2017
- Jianli Chen (Chair)
  Title: “A Case Study of Embedding Real Time Infrastructure Sensor Data to BIM”. Graduated May 2014
- Aruna Muthumanickam (Co-Chair with John Taylor)
  Title: “A framework for integrating energy-use information and BIM using IFCs”. Graduated May 2014

- Ben Watson (Chair)
  Non-Thesis Option. Expected graduation 2026
- Bennett Dziekan (Chair)
  Non-Thesis Option. Expected graduation 2026
- Andrew Bates (Chair)
  Non-Thesis Option. Expected graduation 2026
- Eliza Duncan (Chair)
  Non-Thesis Option. Expected graduation 2026
- Isabella Grena (Chair)
  Non-Thesis Option. Expected graduation 2026
- Sultan Alzahrani (Chair)
  Non-Thesis Option. Expected graduation 2026
- Cassidy Blackmore (Chair)
  Non-Thesis Option. Expected graduation 2026
- Manu Nalamati (Chair)
  Non-Thesis Option. Expected graduation 2026
- Robert Schick (Chair)
  Non-Thesis Option. Expected graduation 2026
- Juan Cundumi Garcia (Chair)
  Non-Thesis Option. Expected graduation 2026

- David Kiser (Chair)
  Non-Thesis Option. Expected graduation 2026
- Kambiz Kia (Chair)
  Non-Thesis Option. Expected graduation 2026
- Meghan Marsh (Chair)
  Non-Thesis Option. Expected graduation 2026
- Veronka Faltes (Chair)
  Non-Thesis Option. Expected graduation 2026
- Yousra Essaadi (Chair)
  Non-Thesis Option. Expected graduation 2026
- Yacine Berrada (Chair)
  Non-Thesis Option. Expected graduation 2026
- Yusna Ayer (Chair)
  Non-Thesis Option. Expected graduation 2026

- Negar (Nina) Pourkhodagholi (Chair)
  Non-Thesis Option. Graduated December 2024
- Ronn Bi Ojes (Chair)
  Non-Thesis Option. Graduated December 2024
- Seerin Siji (Chair)
  Non-Thesis Option. Graduated December 2024
- Yoounes Alkhallaqi (Chair)
  Non-Thesis Option. Graduated December 2024
- Meetkumar Ramnikbhai Bhalala (Chair)
  Non-Thesis Option. Graduated December 2024
- Xiaohan Liu (Chair)
  Non-Thesis Option. Graduated December 2024
- Zachary Luke Meiners (Chair)
  Non-Thesis Option. Graduated May 2024
- Ethan Alexander Boisvert (Chair)
  Non-Thesis Option. Graduated May 2024
- Alexander Russo
  Non-Thesis Option. Graduated December 2023
- Kevin Robinson Jr. (Chair)
  Non-Thesis Option. Graduated December 2023
- John Bennet Miles (Chair)
  Non-Thesis Option. Graduated December 2023
- Kassidie Funk (Chair)
  Non-Thesis Option. Graduated December 2023
- Emily Carroll (Chair)
  Non-Thesis Option. Graduated December 2023
- Coleton Beck (Chair)
  Non-Thesis Option. Graduated December 2023
- Payal Desai (Chair)
  Non-Thesis Option. Graduated December 2023

- Seth Tyler Martin (Committee Member)

Non-Thesis Option. Graduated December 2024

- David Manning Wilson (Committee Member)
  Non-Thesis Option. Graduated December 2019
- Armin Yeganeh (Committee Member)
  Title: “Correlation of Ambient Air Temperature and Cognitive Performance: A Systematic Review and Meta-Analysis”. Graduated October 2019
- Sergio Ballivan (Committee Member)
  Title: “Anonymous Indoor Positioning System using Depth Sensors for Context-Aware Human-Building Interaction”. February 2019
- Dan Moore (Committee Member)
  Title: “Enhanced Structural Steel Supply Chain Management through RFID Technology, SQL Database and Enhanced Model Visualization”. Graduated May 2017
- Suduck Kim (Committee Member)
  Title: “The Difference in BIM Data Requirements Between Normative Representations and Actual Practices” Expected graduation August 2015
- Tugce Duzener (Committee Member)
  Title: “Energy Conservation with Eco-Feedback Systems and its Potential Impact on LEED Certification Systems” Expected graduation August 2015
- Hanh Truong (Committee Member)
  Title: TBD. Expected graduation August 2016
- Parisa Nikkhoo (Committee Member)
  Title: “Adoption of Innovative Products in the US Housing Industry: Builders’ Practices 2000-2010”. Graduated August 2014
- David Dooper (Committee Member)
  Title: “Exploring the Opportunities RFID may provide in the Prevention of Ladder Falls on Construction Sites”. Graduated December 2013
- Brett Medsker (Committee Member)
  Title: “USACE Small Project Delivery Practices”. Graduated May 2013
- Ashley Drake (Committee Member)
  Title: “The Confusion of Construction Education Choices: How does a Masters Student Decide?”. Graduated December 2012
- Viveka Nalluri (Committee Member)
  Title: “Sustainability Analysis for Light Industrial Facilities: Phase 1 - Facility Opportunity Assessment” Graduated August 2012